

the same.

In the digital world, there are two types of filters: FIR filters and IIR filters.

FIR filters. These filters are called finite impulse response filters. They are characterised by the impulse response given as:

$$H(z) = \sum_{n=0}^{N-1} x(n)z^{-n}$$

where the impulse response $H(z)$ is of finite length 'N' and $x(n)$ is the input sequence.

IIR filters. These filters are called infinite impulse response filters. They are characterised by the impulse response given as:

$$H(z) = \frac{\sum_{n=0}^{N_1-1} y(n)z^{-n}}{\sum_{n=0}^{N_2-1} x(n)z^{-n}}$$

where the impulse response $H(z)$ is infinite in duration.

FIR filters can be obtained using truncation of IIR response with suitable window functions. Here we make use of a rectangular window function given as $w(n)=1$. So the desired FIR is obtained as:

$$h_d(n) = \begin{cases} h_d(n)w(n) & 0 \leq n \leq N-1 \\ 0 & \text{otherwise} \end{cases}$$

where $h_d(n)$ is an infinite-duration impulse response.

Software program

There are different methods of designing filters and so are the ways of implementing them. Here we have developed a 'C++'-based software to design FIR filters using rectangular window method and obtain the frequency response of the same. The

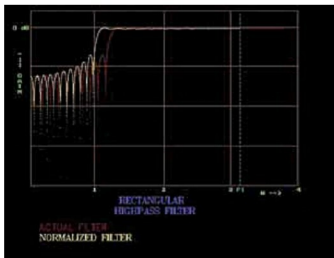


Fig. 3: Frequency response of high-pass filter with $N=61$ and $\omega=1.2$

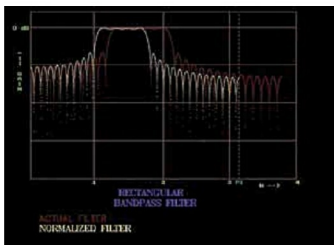


Fig. 4: Frequency response of band-pass filter with $N=61$, $\omega_1=1.2$ and $\omega_2=2.1$

resultant software:

1. Displays a window with a list of various filters.
2. Prompts you to enter the filter type.
3. Allows you to choose the window length (N) up to 200.
4. Prompts you to enter the cut-off frequency in radians (user should enter the values between 0.2 and 2.6).

With this information, the software designed in C++ gives normalised filter response and actual filter response. Normalised filter response is designed

to pass through the cut-off frequency of one radian. Actual filter response is designed to pass through a cut-off frequency specified by the user.

Software structure.

The program is developed using main.cpp, rec.h, graph.h and func.h files. The default path of the program is C:\TC\BIN\FIR\ . The software is designed to follow the structure shown in Fig. 1.

The program files are described below:

main.cpp. This file contains the menu-driven program for user inputs.

func.h. It is a header file containing code related to rectangular window coefficients calculation.

rec.h. It is a header file containing code for supporting C++ graphics.

graph.h. It is a header file again to support graphic content in C++.

Output response. Now let's run the program for different window lengths 'N' to observe the effect of 'N' on frequency response. The output frequency response is the graph of 'N' against the cut-off frequency (ω) expressed in radian per second. A normalised frequency scale is used.

Low-pass filter. Simulation of the low-pass filter using a rectangular window is done by the software. Figs 2(a), (b) and (c) show the outputs of the



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